Question Paper

Physics Single Correct (Maximum Marks: 80)

Question No. 1

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

When $p-n$ junction diode is forward biased, then :

A. the depletion region is reduced and barrier height is increased

B. the depletion region is widened and barrier height is reduced

C. both the depletion region and barrier height are reduced

D. both the depletion region and barrier height are increased

Question No. 2

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Energy levels A, B and C of a certain atom correspond to increasing values of energy, i.e. $E_A < E_B < E_C$. If λ_1, λ_2 and λ_3 are the wavelengths of radiations corresponding to transitions C to B, B to A and C to A respectively, which of the following relations is correct?

A. $\lambda_3 = \lambda_1 + \lambda_2$

B. $\lambda_3 = \frac{\lambda_1 \lambda_2}{\lambda_1 + \lambda_2}$

C. $\lambda_3 = \sqrt{\lambda_1 \lambda_2}$

D. $\lambda_3 = \sqrt{\lambda_1^2 + \lambda_2^2}$

JOIN - @IIT_JEE_NEET_BOOK
u/wrapthatass

Question No. 3

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The energy that should be added to an electron to reduce its de-Broglie wavelength from 1 nm to 0.5 nm is

A. four times the initial energy.

B. equal to the initial energy.

C. twice the initial energy.

D. thrice the initial energy.

Question No. 4

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Assertion: Gamma rays are more energetic than X-rays.
Reason: Gamma rays are of nuclear origin but X- rays are produced due to sudden deceleration of high energy electrons while falling on a metal of high atomic number.

A. both Assertion and Reason are true and the Reason is correct explanation of the Assertion.

B. both Assertion and Reason are true but Reason is not correct explanation of the Assertion.

C. Assertion is true but Reason is false.

D. Assertion is false but Reason is true.

JOIN @IIT_JEE_NEET_BOOK

Question No. 5

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The binding energy per nucleon for deuterons ($^{2}_{1}\mathrm{H}$) and helium ($^{4}_{2}\mathrm{He}$) are 1.1 MeV and 7.0 MeV respectively. The energy released when two deuterons fuse to form a helium nucleus ($^{4}_{2}\mathrm{He}$) is :

A. 28 MeV

B. 2.2 MeV

C. 23.6 MeV

D. 18.6 MeV

Question No. 6

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The correctness of velocity of an electron moving with velocity $50\ \mathrm{ms}^{-1}$ is 0.005%. The accuracy with which its position can be measured will be

A. $4634 \times 10^{-3}\mathrm{m}$

B. $4634 \times 10^{-5}\mathrm{m}$

C. $4634 \times 10^{-6}\mathrm{m}$

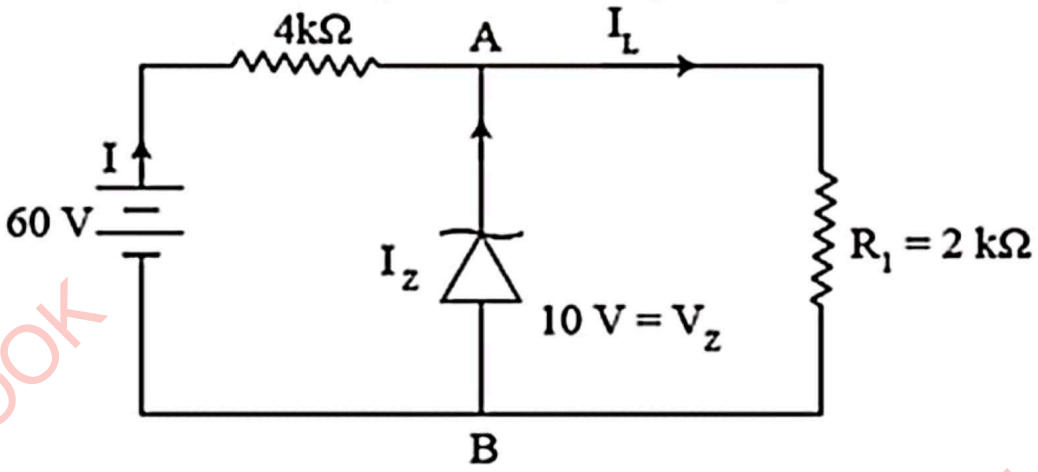
D. $4634 \times 10^{-8}\mathrm{m}$

Question No. 7

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A Zener diode is connected to a battery and a load as shown below: The currents I, I_Z and I_L are respectively



A. 12.5 mA, 7.5 mA, 5 mA

B. 15 mA, 7.5 mA, 7.5 mA

C. 12.5 mA, 5 mA, 7.5 mA

D. 15 mA, 5 mA, 10 mA

Question No. 8

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

In a H-atom spectrum λ_{3-1} and λ_{2-1} represents wavelengths emitted due to transition from second and first excited states to the ground state respectively. The value of $\frac{\lambda_{3-1}}{\lambda_{2-1}}$ is :

A. $27/32$

B. $32/27$

C. 20

D. None of these

JOIN - @IIT_JEE_NEET_BOOK

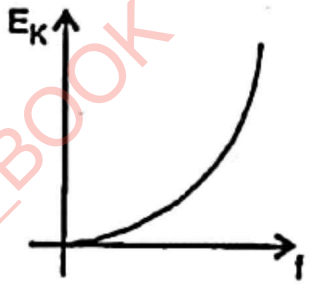
Question No. 9

Only One Option Correct Type

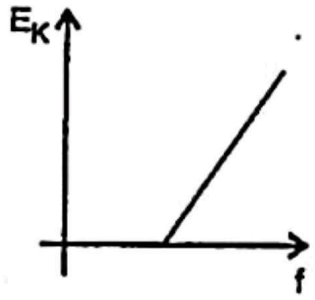
Each question has multiple options out of which ONLY ONE is correct.

Maximum kinetic energy of a photoelectric varies with the frequency (f) of the incident radiation as

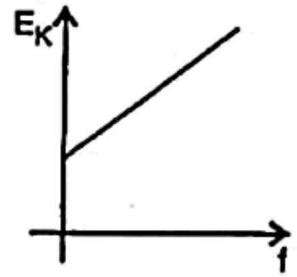
A.



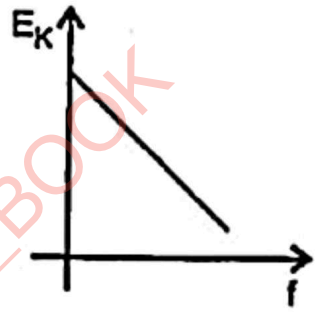
B.



C.



D.



JOIN - @IIT_JEE_NEET_BOOK

Question No. 10

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

When a U^{238} nucleus originally at rest, decays by emitting an alpha particle having a speed ' u ', the recoil speed of the residual nucleus is

A. $\frac{4u}{238}$

B. $-\frac{4u}{234}$

C. $\frac{4u}{234}$

D. $-\frac{4u}{238}$

Question No. 11

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

When 24.8 keV X-rays strike a material, the photoelectrons emitted from K shell are observed to move in a circle of radius 23 mm in a magnetic field of 2×10^{-2} T. The binding energy of K shell electrons is

A. 6.2 keV

B. 5.4 keV

C. 7.4 keV

D. 8.6 keV

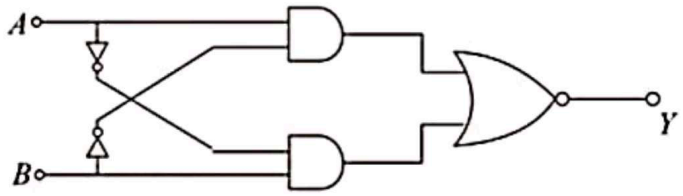
JOIN @IIT_JEE_NEET_BOOK

Question No. 12

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The truth table for the following logic circuit is :



A.

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

B.

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	0

C.

A	B	Y
0	0	0
0	1	1
1	0	0
1	1	1

D. None of the above

Question No. 13

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A pulse of light of duration 100 ns is absorbed completely by a small object initially at rest. Power of the pulse is 30 mW and the speed of light is 3×10^8 ms^{-1}. The final momentum of the object is

A. 0.3×10^{-17} kg ms^{-1}

B. 1.0×10^{-17} kg ms^{-1}

C. 3.0×10^{-17} kg ms^{-1}

D. 9.0×10^{-17} kg ms^{-1}

Question No. 14

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The wavelength of the first spectral line of sodium is 5896 Å. The first excitation potential of sodium atom will be ($h = 6.63 \times 10^{-34}$ J s)

A. 4.2 V

B. 3.5 V

C. 2.1 V

D. None of these

JOIN - @IIT_JEE_NEET_BOOK

Question No. 15

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Choose the incorrect statement.

A. The nuclear force becomes weak if the nucleus contains too many protons compared to the number of neutrons

B. The nuclear force becomes weak if the nucleus contains too many neutrons compared to the number of protons

C. Nuclei with atomic number greater than 82 show a tendency to disintegrate

D. The nuclear force becomes very strong if the nucleus contains a large number of nucleons

JOIN - @IIT_JEE_NEET_BOOK

Question No. 16

Only One Option Correct Type

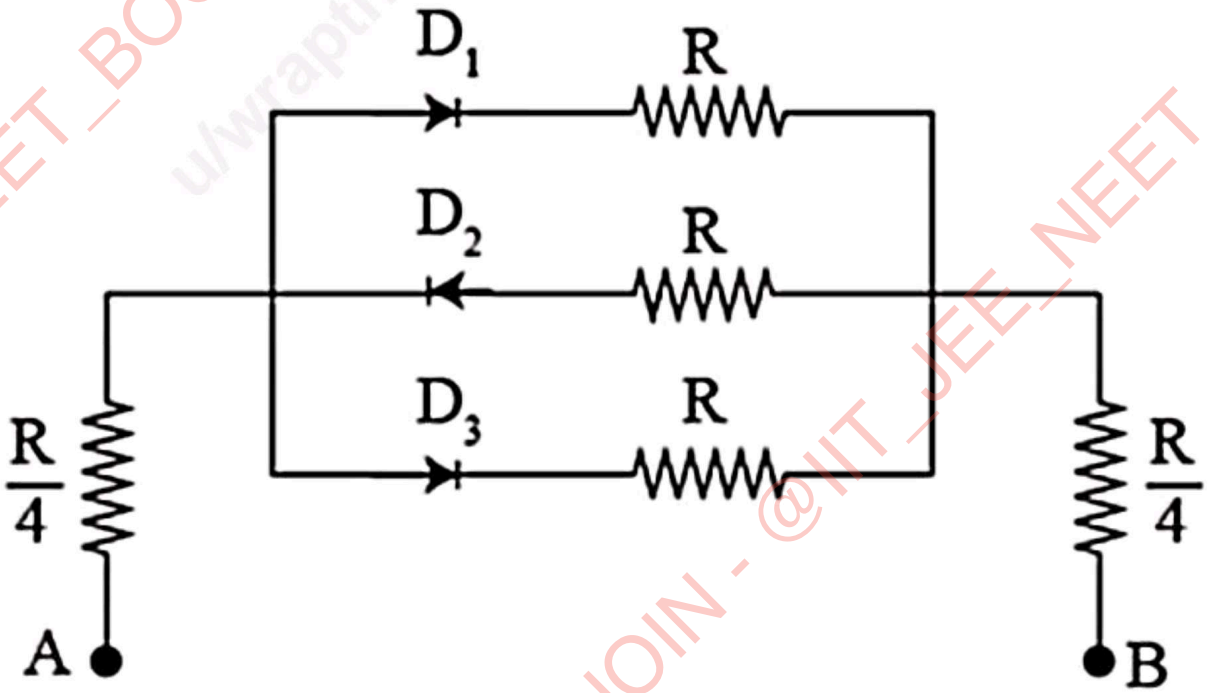
Each question has multiple options out of which ONLY ONE is correct.

In the following circuits, PN-junction diodes D_1, D_2 and D_3 are ideal for the following potential of A and B, the correct increasing order of resistance between A and B will be

(i) -10 V, -5 V

(ii) -5 V, -10 V

(iii) -4 V, -12 V



A. (i) < (ii) < (iii)

B. (iii) < (ii) < (i)

C. (ii) = (iii) < (i)

D. (i) = (iii) < (ii)

Question No. 17

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The electric displacement current through a surface area S is proportional to the

A. Magnetic flux through S

B. Rate of change of magnetic flux through S

C. Electric flux through S

D. Rate of change of electric flux through S

Question No. 18

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If the work functions of three photosensitive materials are 1 eV, 2 eV and 3 eV respectively, then the ratio of the respective frequencies of light that produce photoelectrons of the maximum kinetic energy of 1 eV from each of them is

A. $1 : 2 : 3$

B. $2 : 3 : 4$

C. $1 : 1 : 1$

D. $3 : 2 : 1$

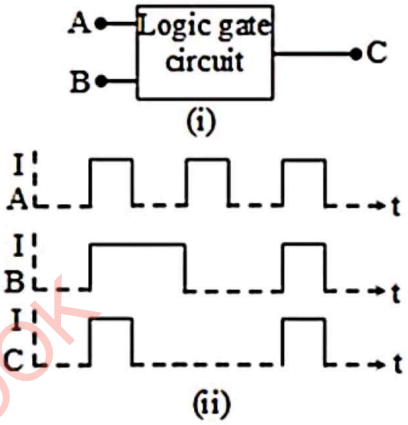
JOIN - @IIT_JEE_NEET_BOOK
www.apthatass

Question No. 19

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The following figure shows a logic gate circuit with two inputs A and B and output C. The voltage waveforms of A, B and C are as shown in second figure below



The logic circuit gate is

A. OR gate

B. AND gate

C. NAND gate

D. NOR gate

Question No. 20

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The rms value of the electric field of the light coming from the Sun is 720 N/C. The average total energy density of the electromagnetic wave is $\left(\varepsilon = 8.85 \times 10^{-12}\right)$

A. $6.37 \times 10^{-9}\ \mathrm{J/m^3}$

B. $81.35 \times 10^{-12}\ \mathrm{J/m^3}$

C. $3.3 \times 10^{-3}\ \mathrm{J/m^3}$

D. $4.58 \times 10^{-6}\ \mathrm{J/m^3}$

JOIN - @IIT_JEE_NEET_BOOK

Physics Numerical (Maximum Marks: 20)

Question No. 1

Numerical Type

The answer has to be filled into the input box provided below.

Wavelengths belonging to Balmer series lying in the range of 450 nm to 750 nm were used to eject photoelectrons from a metal surface whose work function is 2.0eV. The maximum kinetic energy of emitted photoelectron is k eV, find value of 100k and roundoff to nearest integer. (Take hc = 1242eVnm.)

Question No. 2

Numerical Type

The answer has to be filled into the input box provided below.

A hydrogen-like atom of atomic number Z is in an excited state of quantum number $2n$. It can emit a maximum energy photon of 204eV. If it makes a transition to the quantum state n, a photon of energy 40.8eV is emitted. Calculate the atomic number Z. Ground state energy of hydrogen atom is -13.6eV

Question No. 3

Numerical Type

The answer has to be filled into the input box provided below.

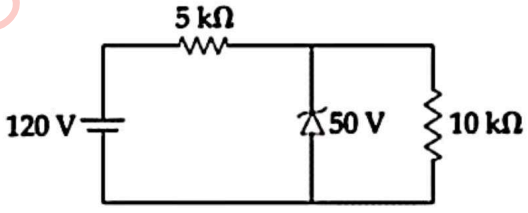
The ratio of the nuclear radius, of an atom with mass number A and ${}_2^4\mathrm{He}$ is $(14)^{1/3}$. What is the value of A?

Question No. 4

Numerical Type

The answer has to be filled into the input box provided below.

For the circuit shown below, the current (in mA) through the Zener diode is



JOIN - @IIT_JEE_NEET_BOOK

Question No. 5

Numerical Type

The answer has to be filled into the input box provided below.

Consider the following nuclear reaction ${}_1H^2 + {}_1H^3 \rightarrow 2He^4$. If 20 meV of energy is released per fusion reaction. Mass of ${}_1H^2$ consumed per day is 0.1 gm. What is the power of the reactor in MW?

Chemistry Single Correct (Maximum Marks: 80)

Question No. 1

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The effect of lanthanoid contraction in the lanthanoid series of elements by and large means

A. increase in atomic radii and decrease in ionic radii,

B. decrease in atomic radii and increase in ionic radii.

C. increase in both atomic and ionic radii.

D. decrease in both atomic and ionic radii.

JOIN - @IIT_JEE_NEET_BOOK

Question No. 2

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A salt on treatment with dil. HCl gives a pungent smelling gas and a yellow precipitate. The salt gives green flame when tested. The salt solution gives a yellow precipitate with potassium chromate. The salt is

A. $NiSO_4$

B. BaS_2O_3

C. PbS_2O_3

D. $CuSO_4$

Question No. 3

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The boiling points of the following hydrides follow the order

A. $SbH_3 > NH_3 > AsH_3 > PH_3$

B. $NH_3 > PH_3 > AsH_3 > SbH_3$

C. $NH_3 > SbH_3 > AsH_3 > PH_3$

D. $SbH_3 > AsH_3 > NH_3 > PH_3$

JOIN @IIT_JEE_NEET_BOOK
u/wrapthatass

Question No. 4

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Which is not true about borax ?

A. It is a useful primary standard for titrating against acids

B. One mole of borax can be used as a buffer

C. Aqueous solution of borax can not be used as buffer

D. It is made up of two triangular BO_3 units and two tetrahedral BO_4 units

Question No. 5

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The INCORRECT statement among the following is _____.

A. C_{60} is an allotropic form of carbon

B. O_3 is an allotropic form of oxygen

C. S_8 is the only allotropic form of sulphur

D. red phosphorus is more stable in air than white phosphorus

JOIN - @IIT_JEE_NEET_BOOK

Question No. 6

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A solution of a metal ion when treated with KI gives a red precipitate which dissolves in excess KI to give a colourless solution. Moreover, the solution of metal ion on treatment with a solution of cobalt (II) thiocyanate gives rise to a deep blue crystalline precipitate. The metal ion is

A. Pb^{2+}

B. Hg^{2+}

C. Cu^{2+}

D. Co^{2+}

Question No. 7

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Match the Columns I and II and select the codes accordingly.

	Column I		Column II
A.	Obtained through Solvay process	I.	K_2CO_3
B.	Green colour due to $[Cr(H_2O)_6]^{3+}$ ion	II.	$S_2O_3^{2-} + Fe$
C.	Green colouration	III.	$SO_3^{2-} + K_2C$
D.	Melts at 850°C	IV.	Na_2CO_3

JOIN - @IIT_JEE_NEET_BOOK

7

Only One Option Correct Type

estion has multiple options out of which ONLY ONE is correct.

Columns I and II and select the codes accordingly.

lumn I		Column II
tained through Solvay process	I.	K_2CO_3
een colour due to $[Cr(H_2O)_6]^{3+}$ ion	II.	$S_2O_3^{2-}$ + $FeCl_3$ solution
een colouration	III.	SO_3^{2-} + $K_2Cr_2O_7/H^+$
lts at 850°C	IV.	Na_2CO_3

	B	C	D
	III	I	II

	B	C	D
	I	III	II

	B	C	D
	III	II	I

B	C	D
I	IV	III

JOIN - @IIT_JEE_NEET_BOOK

A.

A	B	C	D
IV	III	I	II

B.

A	B	C	D
IV	I	III	II

C.

A	B	C	D
IV	III	II	I

D.

A	B	C	D
II	I	IV	III

Question No. 8

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Select the correct statement(s) among following.

A. Al forms AlF_6^{3-}, an octahedral complex

B. Be and Al show diagonal relationship

C. Be forms tetrahedral complex $[Be(C_2O_4)_2]^{2-}$

D. All of the above

JOIN - @IIT_JEE_NEET_BOOK

Question No. 9

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Separation of lanthanides from their mixture is not easy because of

A. Shielding effect

B. Penetrating effect

C. Consequences of lanthanide contraction

D. Inert pair effect

Question No. 10

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Which of the following statements is correct?

A. NO_2 on reaction with NaOH gives a mixture of two salts.

B. N_2O also known as laughing gas, is angular in shape.

C. NO_2 is a sweet smelling and is angular in shape.

D. NO is a colourless gas and acidic in nature.

JOIN - @IIT_JEE_NEET_BOOK

Question No. 11

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Arrange the following in the correct order of Lewis acidity.

$BF_3,\ BCl_3,\ BBr_3$

A. $BF_3 > BBr_3 > BCl_3$

B. $BF_3 > BCl_3 > BBr_3$

C. $BF_3 < BCl_3 < BBr_3$

D. $BBr_3 < BF_3 < BCl_3$

Question No. 12

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

On performing a borax-bead test with a given inorganic mixture for qualitative analysis, the colour of the bead was found to be emerald green both in oxidising and reducing flame. It indicates the possibility of the presence of

A. Co^{2+}

B. Ni^{2+}

C. Cr^{3+}

D. Cu^{2+}

JOIN - @IIT_JEE_NEET_BOOK

Question No. 13

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

CrO_4^{2-} (yellow) changes to $Cr_2O_7^{2-}$ (orange) in pH = x and vice-versa in pH = y. Hence, x and y are

A. 6, 8

B. 6, 5

C. 8, 6

D. 7, 7

Question No. 14

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Nitrogen exist as N_2 but phosphorus exist as P_4 this is due to

A. triple bond exist between phosphorus atom

B. $p_\pi - p_\pi$ bond is weak in P_4

C. $p_\pi - p_\pi$ bond is strong in P_4

D. multiple bonds form easily

JOIN - @IIT_JEE_NEET_BOOK

Question No. 15

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Assertion: La (Z = 57) belongs to the d-block and is an element of group 3 of the periodic table.
Reason: All Lanthanoids belong to the group 3 of the periodic table.

A. Both the assertion and reason statements are true and the reason is the correct explanation of the assertion

B. Both assertion and reason statements are true but the reason is not the correct explanation of the assertion.

C. The assertion statement is true but the reason statement is false

D. Both the assertion and reason statements are false

Question No. 16

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Which of the following halogen acids is least basic?

A. HF

B. HCl

C. HBr

D. HI

JOIN - @IIT_JEE_NEET_BOOK

Question No. 17

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Pick out the correct statement set for transition metals
1. They form interstitial compounds
2. 5d-elements have higher energies than 3 d or 4 d -elements
3. Zr and Hf have almost identical atomic and ionic radii
4. They have low melting and boiling points (or low enthalpies of atomization)

A. 1,2 only

B. 2,3 only

C. 1, 2, 3 only

D. 1, 2, 3, 4

Question No. 18

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Metal chloride A is soluble in hot water but insoluble in cold water. Select correct statement about A. Thus

A. A can give yellow ppt. with K_2CrO_4

B. A can give white ppt with K_2SO_4

C. A can give yellow ppt with KI

D. All of the above are correct statements.

JOIN - @IIT_JEE_NEET_BOOK

Question No. 19

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Compounds A and B are treated with dil. HCl separately. The gas liberated are Y and Z respectively. Y turns acidified $K_2Cr_2O_7$ paper green while Z turns lead acetate paper black. The compounds A and B are respectively.

A. NaCl and Na_2CO_3

B. Na_2SO_3 and Na_2S

C. Na_2S and Na_2SO_3

D. Na_2SO_3 and Na_2SO_4

Question No. 20

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The correct decreasing order of electropositive character among the following elements is :
Fe, Sc, Rb, Br, Te, F, Ca

A. Fe > Sc > Rb > Br > Te > F > Ca

B. Ca > Rb > Sc > Fe > Te > F > Br

C. Rb > Ca > Sc > Fe > Br > Te > F

D. Rb > Ca > Sc > Fe > Te > Br > F

Chemistry Numerical (Maximum Marks: 20)

Question No. 1

Numerical Type

The answer has to be filled into the input box provided below.

Amongst the following, the maximum number of compounds which do not behave as Lewis acids are:
$SnCl_2$, H_3BO_3, $AlCl_3$, CF_4, SiF_4, CCl_4, BF_3 , $SnCl_4$

Question No. 2

Numerical Type

The answer has to be filled into the input box provided below.

The oxidation number of Mn in the product of alkaline oxidative fusion of MnO_2 is

Question No. 3

Numerical Type

The answer has to be filled into the input box provided below.

How many of the following trends is(are) *CORRECT* with respect to properties of hydrides of group 16 elements?
i. Bond angle: $H_2O > H_2S > H_2Se > H_2Te$
ii. Volatility: $H_2O < H_2S > H_2Se > H_2Te$
iii. Thermal stability: $H_2O > H_2S > H_2Se > H_2Te$
iv. Acidic strength: $H_2O > H_2S > H_2Se > H_2Te$

Question No. 4

Numerical Type

The answer has to be filled into the input box provided below.

How many among the following imparts colour to the Bunsen flame, when flame test is carried out?
Ba, Mg, Ca, Be, Na, Sr, K

JOIN - @IIT_JEE_NEET_BOOK

Question No. 5

Numerical Type

The answer has to be filled into the input box provided below.

In the compound $Na_2\left[B_4O_5(OH)_4\right] \cdot 8H_2O$, if the:

(i) number of $B - O - B$ bonds is x.

(ii) number of $B - B$ bonds is y.

(iii) number of sp^2 hybridized B atoms is z. Calculate the value of $x + y + z$.

Mathematics Single Correct (Maximum Marks: 80)

Question No. 1

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The shortest distance between the line $x = y = z$ and the line of intersection of $2x + y + z - 1 = 0$ and $3x + y + 2z - 2 = 0$ is

A. $\frac{1}{\sqrt{2}}$ units

B. $\frac{1}{\sqrt{3}}$ units

C. $\frac{1}{\sqrt{4}}$ units

D. $\frac{1}{\sqrt{5}}$ units

Question No. 2

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If A, B and C are three points position vectors a, b and c respectively, then perpendicular distance of A from the line joining B and C is

A. $\frac{|a\times b\times c|}{2(b\times c)}$

B. $\frac{|a\times b+b\times c+c\times a|}{2|(b-c)|}$

C. $\frac{|a\times b+b\times c+c\times a|}{|(b-c)|}$

D. None of these

Question No. 3

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A line parallel to $6\hat{i}-3\hat{j}+2\hat{k}$ intersects the line $\frac{x}{1}=\frac{y}{2}=\frac{z}{2}$ at A and the line $\frac{x+1}{1}=\frac{y+2}{2}=\frac{z+3}{2}$ at B, then the length of AB is equal to

A. $\frac{3}{\sqrt{7}}$ units

B. $\frac{5}{\sqrt{439}}$ units

C. $\frac{7}{\sqrt{85}}$ units

D. $\frac{9}{\sqrt{425}}$ units

JOIN - @IIT_JEE_NEET_BOOK

Question No. 4

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let $\vec{p} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{r}$ be a variable vector such that $\vec{r} \cdot \hat{i}$, $\vec{r} \cdot \hat{j}$ and $\vec{r} \cdot \hat{k}$ are even natural numbers. If $\vec{r} \cdot \vec{p} \leq 20$, then the number of values of $\vec{r}$ is

A. 20

B. 60

C. 75

D. 120

Question No. 5

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let $V_1 =$ variance of $\{13,16,19.\ldots\ldots\ldots 103\}$ and $V_2 =$ variance of $\{20,26,32.\ldots\ldots\ldots 200\}$. Then $V_1 : V_2$ is

A. $1:2$

B. $1:1$

C. $4:9$

D. $1:4$

JOIN - @IIT_JEE_NEET_BOOK

Question No. 6

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let ΔOAB be an equilateral triangle with side length unity (O being the origin). Also, M and N are the points of trisection of AB, M being closer to A and N being closer to B. Position vectors of A, B, M and N are $\vec{a}, \vec{b}, \vec{m}$ and $\vec{n}$ respectively, then the value of $\vec{m} \cdot \vec{n}$ is equal to

A. $\frac{1}{2}$

B. $\frac{2}{3}$

C. $\frac{13}{18}$

D. $\frac{4}{9}$

Question No. 7

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The length of the perpendicular (in units) from the point $(1,2,4)$ on the straight line $\frac{x-2}{1} = \frac{y-7}{2} = \frac{z-3}{-1}$ lies in the interval

A. $\left(1, \frac{3}{2}\right)$

B. $(2,3)$

C. $(0,2]$

D. $[4,5)$

JOIN @IIT_JEE_NEET_BOOK

Question No. 8

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let $A(4, 7, 8)$, $B(2, 3, 4)$ and $C(2, 5, 7)$ be the position vectors of the vertices of a ΔABC. The length of the internal bisector of the angle of A is

A. $\frac{3}{2}\sqrt{34}$

B. $\frac{2}{3}\sqrt{34}$

C. $\frac{1}{2}\sqrt{34}$

D. $\frac{1}{3}\sqrt{34}$

Question No. 9

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The direction cosines l, m and n of two lines are connected by the relations $l + m + n = 0$ and $lm = 0$, then the angle between the lines is

A. $\frac{\pi}{3}$

B. $\frac{\pi}{4}$

C. $\frac{\pi}{2}$

D. 0

JOIN - @IIT_JEE_NEET_BOOK

Question No. 10

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If $x_1, x_2, x_3 \ldots x_{34}$ are numbers such that
$x_i = x_{i+1} = 150, \forall\, i \in \{1, 2, 3, 4, \ldots .9\}$ and
$x_{i+1} - x_i = -2, \forall\, i \in \{10, 11, \ldots .33\}$, then median of
$x_1, x_2, x_3 \ldots \ldots x_{34}$ is

A. 134

B. 135

C. 148

D. 150

Question No. 11

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let $\overrightarrow{x}, \overrightarrow{y}$ are non zero and non collinear vectors such that
$\left(a\alpha^2 + b\alpha + c\right)\vec{x} + \left(a\beta^2 + b\beta + c\right)\vec{y} + \left(a\gamma^2 + b\gamma + c\right)(\vec{x} \times \vec{y}) = 0,$
then $(a + b + c)$ is (where $a, b, c \in R$)

A. -2

B. -1

C. 0

D. 4

JOIN - @IIT_JEE_NEET_BOOK
u/wrapthatass

Question No. 12

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If the lines $\frac{x-1}{1}=\frac{y-3}{1}=\frac{z-2}{\lambda}$ and $\frac{x-1}{\lambda}=\frac{y-3}{2}=\frac{z-4}{1}$ intersect at a point, then the value of λ^2+4 is equal to

A. 8

B. 10

C. 13

D. 5

Question No. 13

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Coefficient of variation of two distributions are 60% and 75%, and their standard deviation are 18 and 15 respectively, then their arithmetic means respectively are

A. 30, 30

B. 30, 20

C. 20, 30

D. 20, 20

JOIN - @IIT_JEE_NEET_BOOK

Question No. 14

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The line passing through the points $(5, 1, a)$ and $(3, b, 1)$ crosses the $yz-$ plane at the point $\left(0, \frac{17}{2}, \frac{-13}{2}\right)$. Then

A. $a = 2, b = 8$

B. $a = 4, b = 6$

C. $a = 6, b = 4$

D. $a = 8, b = 2$

Question No. 15

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Consider a line passing through the point $\mathrm{P}(3, 4, 5)$& having direction ratios $(2, -2, 1)$. If image of point $\mathrm{Q}(1, -1, 8)$ w.r.t. given line is Q', then length of median of $\triangle \mathrm{PQQ}'$ through vertex P is -

A. 2

B. 3

C. 2.5

D. 3.5

JOIN - @IIT_JEE_NEET_BOOK

Question No. 16

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If the points with position vectors $\alpha\hat{i}+10\hat{j}+13\hat{k}$, $6\hat{i}+11\hat{j}+11\hat{k}$, $\frac{9}{2}\hat{i}+\beta\hat{j}-8\hat{k}$ are collinear, then $(19\alpha-6\beta)^2$ is equal to

A. 36

B. 25

C. 49

D. 16

Question No. 17

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If $\vec{a}=\hat{i}-2\hat{j}+3\hat{k}$ and $\vec{b}=-3\hat{i}+\hat{j}-\hat{k}$ and $\vec{r}\times\vec{a}=\vec{b}\times\vec{a}$, $\vec{r}\times\vec{b}=\vec{a}\times\vec{b}$ then a unit vector in the direction of $\vec{r}$ is

A. $\frac{1}{3}(-2\hat{i}+\hat{j}-\hat{k})$

B. $\frac{1}{3}(-2\hat{i}-\hat{j}+2\hat{k})$

C. $\frac{1}{3}(-2\hat{i}-\hat{j}-2\hat{k})$

D. None of these

Question No. 18

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The direction cosines of two lines satisfy $2l + 2m - n = 0$ and $lm + mn + nl = 0$. The angle between these lines is

A. $\frac{\pi}{4}$

B. $\frac{\pi}{2}$

C. $\frac{\pi}{6}$

D. $\frac{\pi}{3}$

Question No. 19

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let $\vec{a} = 3\hat{i} + 2\hat{j} + 2\hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} - 2\hat{k}$ be two vectors. If a vector perpendicular to both the vectors $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$ has the magnitude 12 then one such vector is:

A. $4\left(2\hat{i} + 2\hat{j} + \hat{k}\right)$

B. $4\left(2\hat{i} - 2\hat{j} - \hat{k}\right)$

C. $4\left(-2\hat{i} - 2\hat{j} + \hat{k}\right)$

D. $4\left(2\hat{i} + 2\hat{j} - \hat{k}\right)$

JOIN @IIT_JEE_NEET_BOOK

Question No. 20

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The perpendicular distance of $P(1, 2, 3)$ from the line $\frac{x-6}{3} = \frac{y-7}{2} = \frac{z-7}{-2}$ is

A. 7

B. 5

C. 0

D. 6

Mathematics Numerical (Maximum Marks: 20)

Question No. 1

Numerical Type

The answer has to be filled into the input box provided below.

Given two skew-lines with parametric equations $x = 1 + t,\ y = 1 + 6t,\ z = 2t$ and $x = 1 + 2s,\ y = 5 + 15s,\ z = -2 + s$. The shortest distance between these two given lines is equal to

Question No. 2

Numerical Type

The answer has to be filled into the input box provided below.

Mean of 100 observations is 45 . If it was later found that two observations 19 and 31 were incorrectly recorded as 91 and 13 . The correct mean is n. what is value of [n]. ([.] is greatest integer function).

JOIN - @IIT_JEE_NEET_BOOK

Question No. 3

Numerical Type

The answer has to be filled into the input box provided below.

In ΔABC, a point P is chosen on side $\overrightarrow{AB}$ so that $AP : PB = 1 : 4$ and a point Q is chosen on the side $\overrightarrow{BC}$ so that $CQ : QB = 1 : 3$. Segment $\overrightarrow{CP}$ and $\overrightarrow{AQ}$ intersects at M. If the ratio $\frac{MC}{PC}$ is expressed as a rational number in the lowest term as $\frac{a}{b}$, then find $|a - b|$.

Question No. 4

Numerical Type

The answer has to be filled into the input box provided below.

If the length intercepted by a line with direction ratios $2, 7, -5$ between the lines $\frac{x-5}{3} = \frac{y-7}{-1} = \frac{z+2}{1}$ and $\frac{x+3}{-3} = \frac{y-3}{2} = \frac{z-6}{4}$ is $\sqrt{k}$, then $k =$

Question No. 5

Numerical Type

The answer has to be filled into the input box provided below.

Let $|\vec{a}| = 1$, $|\vec{b}| = 4$ and $\vec{a} \times \vec{r} + \vec{b} = \vec{r}$. If the projection of $\vec{r}$ along $\vec{a}$ is 2, then the projection of $\vec{r}$ along $\vec{b}$ is K, find 4K